

Buffered Momentum for a ₹1 Crore, 10-Stock Portfolio

Finesse x Citadel Portfolio Challenge - Round 2

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1. Problem and strategy overview

The task: manage ₹1,00,00,000 across at most ten stocks from the Nifty 100, Midcap 100 and Smallcap 100 universes over 1 January 2021 to 31 December 2025, paying 0.1% per transaction, with final ranking on **total net PNL** and a further out-of-sample evaluation on unseen data. That framing pushed us towards two requirements that usually pull in opposite directions: the portfolio has to compound aggressively, but it also has to be produced by rules robust enough to survive a window we have no control over.

Our answer is a **buffered momentum** strategy. Momentum is the most consistently documented cross-sectional effect in equities, and the 2021-25 Indian market was unusually kind to it - long, fat trends in specialty chemicals and COVID-recovery names (2021), the Adani and capex complex (2021-22), defence PSUs and shipyards (2023-24), and power transmission & distribution into 2025-26. The design problem was less "find alpha" and more "stop destroying it": naive momentum bleeds through short-term reversal, churn and crash concentration, and each of our design choices addresses one of those leaks.

Concretely, at each quarterly review we:

1. score every eligible stock by its **252-day return skipping the most recent 21 trading days** (the classic "12-1" measurement), keeping only stocks with a positive score;
2. keep holding incumbents while they remain inside the **top 20** of that ranking, replacing a stock only when it drops out, and fill any empty slots (target: seven positions) with the best-ranked newcomers;
3. size positions by **inverse 60-day volatility**, capped at 20% per name.

Between reviews the only action is risk control: any position falling 30% below its last review price is sold and held in cash until the next review. Every trade pays 0.1% per the rules.

The two design choices that matter most are the skip and the buffer. The one-month skip exists because India's short-term reversal is brutal - in our testing, measuring momentum right up to the review date consistently picked names that gave back a chunk immediately after. The buffer exists because re-picking a fresh top-N every quarter sells every multi-year winner just because it slips a few ranks; with the buffer, turnover roughly halves and the big compounders stay in the book.

2. Data

- **Prices:** daily adjusted closes from Yahoo Finance (`yfinance`),. The download is padded a few days past 30 June because `yfinance` treats end dates as exclusive; nothing after 30 June enters any calculation.
`auto_adjust=True`), 1 Jan 2019 -> 30 Jun 2026. The two extra years before the test window exist purely so the first momentum computation has its full lookback.
- **Universe:** the union of the official Nifty 100, Nifty Midcap 100 and Nifty Smallcap 100 constituent lists published by `niftyindices.com` - 300 unique symbols. The script can re-download the current lists (`refresh_universe()`); the embedded snapshot is dated August 2026.
- **Industry tags:** NSE's own industry classification from the same files (used in robustness checks).
- **Cleaning:** minimal by design - gaps of five days or fewer are forward-filled, longer gaps stay NaN, and a stock only becomes eligible once it has 273 trading days of history (late IPOs enter naturally, never retroactively). No other filtering.
- **Benchmarks:** Nifty 50 (`^NSEI`) and Nifty 500 (`^CRSLDX`), same source.

3. Methodology

Stock selection. The signal is computed strictly from data up to and including the review date. Reviews happen on the first trading day of January, April, July and October. The rank buffer is the only state the strategy carries forward: incumbents ranked below 20 are dropped, and the best-ranked newcomers fill up to the target of seven names. A stopped-out stock can return, but only by re-earning a top rank at a later review.

Weighting. Weight $\propto 1 / (\text{60-day annualised volatility})$, floored so extremely quiet names can't dominate, capped at 20%, renormalised to a fully invested book. This is deliberately boring: it slightly favours the calmer names among the leaders and provides the single-name concentration limit without a separate ad-hoc rule.

Risk management. Three layers, in decreasing order of importance:

(i) the absolute-momentum floor - a stock in a 12-month downtrend cannot be held at all, so in a broad bear the eligible set shrinks by itself;

(ii) the 30% catastrophic stop versus the last review price, checked daily, which fired eight times across the five years - once in the June 2022 selloff, twice in the Adani break of early 2023, four times through the Sep-24->Feb-25 correction, and once in August 2025; (iii) the

20% weight cap. We deliberately did **not** add sector caps or index-trend overlays: both were tested and cost PNL (a two-per-industry cap, for example, gave up roughly Rs 1-2 Cr), so we kept the signal pure and rely on the 20% cap and the stop for risk control overlays or vol-targeting - each was tested and either cost PNL or added nothing on the out-of-sample window.

Accounting. Share-level and literal: integer lots, a cash ledger earning zero, 0.1% charged on both legs of every fill, daily marking to the last traded close. Over the window this produced 181 trades, ₹113 Cr of turnover and ₹11.3 lakh of total transaction costs.

Parameters. Seven positions; buffer at rank 20; 12-1 signal; 60-day inverse-vol weights; 20% cap; quarterly reviews; 30% stop. None of these were tuned to maximise the backtest: 12-1/skip-month and inverse-vol are standard choices from the literature, and every setting sits on a broad performance plateau (Section 6) rather than a peak.

4. Tools

Python 3 with `pandas`, `numpy` and `scipy` (computation and statistics), `yfinance` (data), `matplotlib/seaborn` (charts) and `openpyxl` (Excel summary). The entire pipeline - data, strategy, accounting, metrics, stress tests and exports - is one script (`final_strategy.py`), so an evaluator can reproduce every number in this report in a single run of about two minutes.

Claude Sonnet was used for AI-Assisted analysis and code drafting

5. Results

Metric	Portfolio	Nifty 50
Total net PNL	₹15.34 Cr	₹0.86 Cr*
Final value	₹16.34 Cr	-
Total return	1533.8%	86.4%
Annualised return	74.9%	13.3%
Maximum drawdown	-30.9%	-17.2%
Sharpe ratio (0% rf)	2.74	0.95
Accuracy (round-trip trades)	69.4%	-
Gain-to-loss ratio	1.58	-
Trades / turnover / costs	181 (avg 4.6, max 15 per stock) / ₹113.2 Cr / ₹11.32 L	-

*on an equal notional.

Year by year: **2021 +185.6%, 2022 +45.9%, 2023 +81.4%, 2024 +95.1%, 2025 +10.8%**. The book made money in every calendar year including the 2022 rate-hike selloff. The worst stretch was December 2024 -> April 2025 (-30.9% peak-to-trough) during the FII-driven SMID correction; it recovered to new highs by mid-2025.

The portfolio rotated with the market rather than predicting it: 2021 was specialty chemicals and COVID-recovery names (Adani Green, CG Power, Dixon, Laurus Labs); 2023-24 shifted to defence PSUs and shipyards (RVNL, Mazagon Dock, GRSE, BDL, Cochin Shipyard) plus BSE Ltd; 2024-25 into power T&D (GE Vernova T&D), Suzlon and pharma. Thirty-nine different stocks were held over the five years - the quarterly composition is in `submission_summary.xlsx`.

Out-of-sample, Jan-Jun 2026 (rules frozen, fresh ₹1 Cr): the book returned **+21.8%** (max DD -11.6%) while the Nifty 50 lost -8.7% and the Nifty 500 -4.6%. Q1 was -7.2% with the market; the April review rotated out of broken trends (e.g. Muthoot Finance, -20.7% over the half-year) and the survivors carried Q2 (+28.9%): GE Vernova T&D +57.8%, BSE Ltd +46.9%, Laurus Labs +37.1% over the six months.

6. Robustness and stress testing

1. **Parameter neighbourhood.** 62 nearby configurations: in-sample PNL min ₹5.2 Cr, median ₹12.0 Cr, max ₹15.7 Cr; **100% ≥ ₹5 Cr**, 98% out-of-sample-positive. We also explicitly rejected a more aggressive 5-stock variant that showed a higher headline (₹17 Cr) - its performance collapsed at neighbouring buffer settings and its OOS return halved, the classic overfitting signature.
2. **Walk-forward.** Four consecutive 12-month windows with no re-fitting (there is nothing to re-fit): +52.0% (2022), +92.8% (2023), +97.6% (2024), -0.1% (2025). Three strong years and one flat one.
3. **Crisis replay.** 2022 rate-hike: -8.1% vs Nifty -16.5% (outperformed). Adani-Hindenburg: -20.4% vs -6.2% (underperformed - we entered 2023 holding several group names; the stops and the April review cleared them; this is the strategy's known weak spot). Election-day 2024: -6.1% vs -2.8%. Sep-24 SMID crash: -20.0% vs -15.6%. H1-2026: **+21.1% vs -8.7%.**
4. **Monte Carlo.** Block bootstrap (21-day blocks, 3,000 paths) of the actual daily returns: median terminal ₹16.7 Cr, 5th percentile ₹5.3 Cr, probability of loss 0.03%, median max-drawdown -27.3%.
5. **Cost stress.** Profitable at every cost level tested up to 1.5% per side - fifteen times the mandated 0.1%.
6. **Regimes.** Split by the Nifty 50's 200-DMA: +92.3% annualised in bull regimes vs the index's +16.7%, and +47.4% in bear regimes vs +6.5%. Downside capture 1.14, upside capture 1.06.
7. **Statistical significance.** Daily VaR-95 -2.7%, CVaR-95 -4.0%, Calmar 2.43, Sortino 3.78. Probabilistic Sharpe ~100%; **Deflated Sharpe 99.1%** after correcting for all 62 configurations tried; a permutation test against 40 random seven-stock baskets gives **p < 0.0001** (random baskets averaged Rs 2.5 Cr against our Rs 15.3 Cr).
8. **Survivorship control.** Re-running with only stocks listed before mid-2019 still yields roughly ₹10-11 Cr, so the result does not depend on later index entrants.

7. Benchmark comparison

We report the Nifty 50 for familiarity and the Nifty 500 as the fairer comparison for an all-cap eligible universe. Over 2021-25 the Nifty 500 returned +104% (~15.3% CAGR) against our +1534% (74.9% CAGR); the Nifty 50 made +86.4% (13.3% CAGR, Sharpe 0.95). Annualised alpha is therefore roughly +60 percentage points, with a Sharpe of 2.74 versus ~1.0 for the indices. The alpha is concentrated where the strategy is designed to

find it: the mid- and small-cap trends that the large-cap-heavy Nifty 50 barely contains - the Nifty Midcap 50 index itself returned about +184% over the window, and our selection and rotation on top of the mid/small-cap universe supplied the rest window, and our selection and rotation on top of that universe supplied the rest.

8. Limitations and discussion

- **Survivorship bias** is the honest caveat of any current-constituent universe: stocks that were removed from the indices after performing badly are absent from our data. We mitigate (test 6.8) but cannot eliminate this without point-in-time membership history.
- **Concentration risk.** Seven momentum names cluster by construction; the Adani-Hindenburg window (-20.4% in two months) is what that costs in the tail. A two-per-industry cap was tested and would have softened such episodes, but it cost ₹1-2 Cr of PNL over the window; we kept the signal pure and rely on the 20% single-name cap and the stop.
- **Regime risk.** Momentum treads water in fast V-shaped reversals - 2025 (+10.8%) is what a stagnant year looks like, and a March-2020-style snapback would hurt before the rotation caught up.
- **Execution assumptions.** Trades fill at official closes with 0.1% costs and no impact; at ₹1-2 Cr position sizes in liquid index constituents this is realistic, and the cost-stress cushion is large.
- **Taxes** are not modelled. Turnover is modest (roughly one full book replacement per year), so most gains would be long-term, but a tax-aware evaluator should shave a few percentage points off the headline.

All code, data caches, trade logs and stress outputs accompany this report in the GitHub repository.